When Voices Speak Louder: Leveraging Audio Signals in Emotion-Cause Extraction via Large Multilingual Multimodal Indian Dialogue Datasets

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***Abstract*—Multimodal emotion and cause recognition have progressed significantly since their inception. Existing multimodal English datasets and newer research extending to languages like Chinese and Polish have been developed to identify conversation emotion-cause pairs. However, these datasets often lack cultural relevance, particularly for Indian languages, which remain underrepresented in the field. This paper aims to bridge this gap by introducing new datasets in India's two most spoken languages: Hindi and Bengali. Our work provides essential resources for more culturally and linguistically diverse emotion and cause recognition tasks, contributing to the broader goal of enhancing detection in multilingual, multimodal Indian contexts.**

***Index Terms*—Benchmark dataset, benglish, emotion cause extraction, hinglish, spoken dialogues.**

TABLE I
EXISTING MULTIMODAL DATASETS FOR EMOTION AND CAUSE DETECTION. *NOTE: EN ← ENGLISH, CN ← CHINESE, IT ← ITALIAN, SP ← SPANISH, PO ← POLISH, HI ← HINDI AND BN ← BENGALI*

	Name	Lang	Modality	Dialogues	Annotations
Only Emotion	IEMOCAP [11]	EN	{v,a,t}	151	10 emotions
	MELD [12]	EN	{v,a,t}	1,433	7 emotions
	EMOTyDA [13]	EN	{v,a,t}	1,341	7 emotions
	M^3ED [14]	CN	{v,a,t}	990	7 emotions
	MEmoR [15]	EN	{v,a,t}	8,536	14 emotions
	FriendsPersona [16]	EN	{v,a,t}	711	Big Five personality traits
	CPED [17]	CN	{v,a,t}	12K	13 emotions
	MSP-IMPROV [18]	EN	{v,a,t}	7,818	11 emotions
	Emo-Film [19]	EN, IT, SP	{v,a,t}	1,115	5 emotions
	RAVDESS [20]	EN	{v,a}	7,356	8 emotions
	nEMO [21]	PO	{a}	4,481	6 emotions
With Cause	ECF [5]	EN	{v,a,t}	1,374	6 emotions
	ECGF [6]	EN	{v,a,t}	1,374	6 emotions
	HinEmo (Ours)	EN+HI	{a,t}	**21K**	7 emotions
	BenEmo (Ours)	EN+BN	{a,t}	**21K**	7 emotions

I. INTRODUCTION

WITH the ever-growing integration of artificial intelligence into daily life tasks, there is an increasing demand for systems that go beyond surface-level language understanding. The natural language processing (NLP) area has emerged successfully to do so with tasks like emotion classification and cause detection. They are aimed at identifying not only the emotions expressed in text but also the triggers behind them. In domains such as human-computer interaction, mental health monitoring, customer service, etc., they are particularly valuable for relating more with the user and understanding them better. Traditionally, emotion and cause detection were largely based on textual data. However, relying solely on text has inherent limitations in fully capturing the richness and complexity of human emotions. As a result, speech emotion recognition (SER) systems[1], [2], [3], [4] and multimodal emotion cause pair extraction[5], [6], [7], [8], [9], [10] gained popularity in grasping the depth of emotional expression.

Most existing research on multimodal emotion and cause detection, along with the datasets developed for these tasks, has focused primarily on English, as summarized in Table I. This presents a significant limitation in multilingual and culturally diverse regions like India, where English is widely spoken but is not the native language for the majority of the population. These datasets are often derived from scripted, monolingual interactions, reducing their effectiveness in real-world Indian scenarios where emotional expression frequently involves a fluid mix of languages, dialects, and culturally grounded communication patterns. Addressing this gap is essential for building more robust and inclusive systems—ranging from voice assistants to emotion-aware moderation tools and sentiment analysis platforms—that can better understand and respond to the way users actually communicate in personal, professional, and digital environments.

To this end, we present the first large-scale multilingual multimodal datasets, **HinEmo** and **BenEmo**, targeting the two top-most spoken languages in India[22]: Hindi and Bengali, respectively. These mixed-language dialogue datasets contain multimodal data (text + audio) with labeled emotions and the corresponding causes at the utterance level. Crucially, they reflect naturalistic, code-switched speech, capturing the linguistic and prosodic features characteristic of Indian multilingual communication. We provide a detailed analysis of the role of audio features in improving cause identification across language pairs. Alongside, we also depict the effectiveness of the datasets by comparing the baselines for three different NLP tasks, namely emotion detection, cause detection, and empathetic response generation. We hope these datasets serve as a stepping stone for developing more inclusive and culturally grounded emotion understanding systems and for advancing research in code-switched and multimodal NLP.

Received 24 April 2025; revised 25 October 2025; accepted 26 October 2025. Date of publication 30 October 2025; date of current version 2 December 2025. The associate editor coordinating the review of this article and approving it for publication was Dr. Arash Mohammadi. *(Nishant Kumar and Srishti Gupta contributed equally to this work.) (Corresponding author: Nishant Kumar.)*

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Digital Object Identifier 10.1109/LSP.2025.3627537

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TABLE II
DATASET STATISTICS

Statistics	HinEmo	BenEmo
# of modality	{Audio,Text}	{Audio,Text}
# of dialogues	21K	21K
# of utterances	69K	67K
# of speakers	45	31
Average audio duration	27 sec	16 sec
Average utterance length	14-15 words	10-11 words
Max utterance length	24-25 words	17-18 words
Max audio length	59 sec	41 sec
Male:Female ratio	3:2	3:2
Age range	18-35 years	18-35 years
Average SNR	27.11dB	29.04dB

II. RELATED WORK

The first work in multimodal emotion data began in 2008 with[11], who introduced the *IEMOCAP* dataset, sparking the field of speech emotion recognition[1], [2], [3], [4]. In 2017,[18] introduced *MSP-IMPROV*, focused on controlling lexical content, emotion, and naturalness. The following year,[20] released *RAVDESS* in English for more ecologically valid emotional representation, while[19] focused on the multilingual *Emo-Film*. In 2019,[12] introduced *MELD*, a large-scale multimodal conversational dataset. By 2020, works like[13], [15], [16], [23] proposed various datasets and models for multimodal emotion detection in conversations. In 2022, [14], [17] targeted the task in Chinese, while[21] addressed it in Polish in 2024. To address the lack of emotion-cause pair targeting in previous works,[5] benchmarked *ECF* dataset for emotion-cause extraction, later enhanced by[6] in 2024. However, none of these datasets capture the linguistic properties of Indian languages, motivating the creation of our datasets.

III. DATASET

In this Section, we describe the methodology adopted to create multilingual multimodal datasets consisting of sample conversations in two languages- HinEmo for Hindi and BenEmo for Bengali. These datasets are intended to help construct efficient models for various emotion and cause detection tasks.

A. Data Collection

To develop robust and culturally relevant models, we construct two large-scale multimodal datasets. A total of 42,000 audio samples were collected (21,000 for HinEmo, 21,000 for BenEmo) in WAV format, ensuring clear audio quality and consistent sampling rates (16 kHz, mono) from diverse sources including: 1) Publicly available speech corpora, 2) Social media content (with proper usage rights) and 3) Emotion-centric audio clips from podcasts, interviews, and online media.

B. Data Preparation

In order to create the datasets, all audio files are first normalized to ensure consistent volume levels and then cleaned to remove any background noise where necessary. As mentioned in Table II, the datasets exhibit good Signal-to-Noise Ratio (SNR), meaning the signal is much stronger than the imperceptible background noise. To achieve multimodality, transcripts were generated for each recording: HinEmo transcripts in Hindi Devanagari script and BenEmo transcripts in Bengali script. The transcription process began with ASR (Automatic Speech Recognition)-assisted transcription using the Whisper API[24], followed by manual verification. The six annotators (three for Hindi, three for Bengali) ensure the accuracy of the transcripts. For the annotator selection, we chose ten research scholars fluent in Hindi and Bengali. Based on their performance in two language proficiency tests—one for Hindi[1] and one for Bengali,[2] the annotators were selected.

C. Data Labelling

This section involves assigning emotion and cause labels to the multimodal dataset over seven core emotions[25]: *Happy*, *Sad*, *Anger*, *Fear*, *Disgust*, *Surprise*, and *Neutral*. To expedite the labeling process of these vast datasets, we follow the approach by[26], utilizing pool-based active learning[27]. Figure 1 illustrates the process for one dataset, which is also applied to the other. Notably, we annotate the audio files first and align the text labels accordingly, leveraging the superior audio-based features demonstrated in prior SER research. The first step involves the manual labeling of a seed dataset (2,000 random files) by all three annotators with one of the seven emotions based on auditory perception. The inter-annotator agreement via Fleiss' Kappa statistic[28] was evaluated to be 0.846, which indicates almost perfect agreement among annotators for labeling. For cases with disagreement, the onboard professionals determined the final label.

The first iteration involves training an initial model for 30 epochs using the manually labeled 2,000 files, to develop a multi-class classification model leveraging a hybrid CNN-Attention-LSTM architecture. The CNN layers act as a feature extractor, focusing on local temporal features (e.g., tone shifts, short-term energy bursts, Mel-frequency cepstral coefficients, pitch, etc). Since not all audio parts contribute equally to emotional content, we highlight the emotionally important features using Attention. Once attention helps emphasize the "hot spots" in the CNN output, the LSTM can better model how the emotion develops over time. The attention after the LSTM weighs the importance of different temporal contexts more globally. We then use this model after iteration 1 to predict 1,000 out of the unlabeled 19,000. The labels with confidence scores < 0.75 are stored for manual verification by human annotators to ensure the accuracy of auto-labeled data. Files with predictions above the threshold and the ones rectified by annotators are used in re-training the model in iteration 2. After re-training, the prediction is then performed on the next 1000 unlabelled files out of 18,000. This iteration is done 7 times with human annotator verification, and then the model reaches a commendable accuracy of 92.3%. This model is then used to predict all remaining 12,000 unlabelled files. The initial threshold of 0.75 was chosen arbitrarily but proved effective after 7 iterations. To enrich the dataset for emotion cause analysis, we utilized both audio and text features to determine the underlying cause of the expressed emotion. Audio features like pitch variations, intonation, etc, and textual features like sentiment keywords, and contextual words like "क्योंकि" (read as kyonki, meaning because in Hindi) were employed to manually assign cause labels.

D. Data Verification

From the 12,000 files, those predicted with confidence scores < 0.75 were verified and rectified by our onboard professionals to ensure the accuracy of the auto-labeled data.

[1]https://indianlingua.com/hindi-level-test/
[2]https://testyourlanguage.com/bengali-quiz

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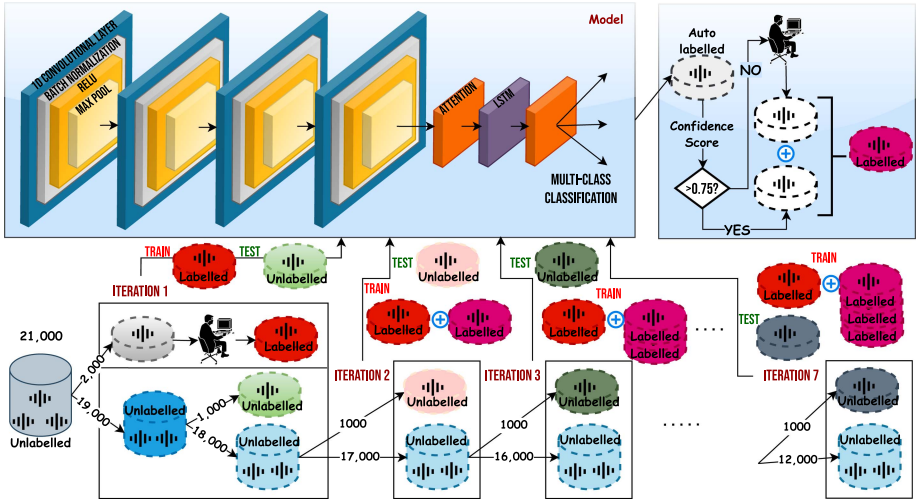

Fig. 1. Block diagram representing the dataset labeling process.

TABLE III
FEATURE SELECTION BASED ON RESULTS IN EACH ITERATION FOR CAUSE EXTRACTION TASK ON BOTH DATASETS. *NOTE: FOLLOWING EXISTING WORKS IN CAUSE EXTRACTION, THE METRICS USED ARE* $P \leftarrow Precision, R \leftarrow Recall, F1 \leftarrow F1-Score,$ AND $A \leftarrow Accuracy$

	Feature	Iteration 1				Iteration 2				Iteration 3				Iteration 4				Iteration 5			
		P	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*
HinEmo	Stress	0.66	0.60	0.61	0.65	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Pause	0.60	0.52	0.63	0.50	0.53	0.61	0.61	0.59	0.67	0.59	0.58	0.70	-	-	-	-	-	-	-	-
	Tone	0.49	0.42	0.47	0.51	0.54	0.61	0.59	0.63	-	-	-	-	-	-	-	-	-	-	-	-
	Pitch	0.38	0.45	0.37	0.34	0.48	0.37	0.45	0.44	0.46	0.36	0.39	0.41	0.43	0.48	0.42	0.49	0.30	0.38	0.45	0.33
	Energy	0.33	0.29	0.34	0.31	0.29	0.32	0.30	0.29	0.34	0.29	0.34	0.34	0.36	0.33	0.33	0.38	0.30	0.29	0.31	0.30
BenEmo	Stress	0.58	0.63	0.55	0.60	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Pause	0.51	0.51	0.49	0.51	0.60	0.64	0.73	0.74	-	-	-	-	-	-	-	-	-	-	-	-
	Tone	0.54	0.44	0.56	0.48	0.44	0.48	0.53	0.59	0.65	0.67	0.68	0.75	-	-	-	-	-	-	-	-
	Pitch	0.42	0.47	0.41	0.29	0.43	0.42	0.38	0.42	0.50	0.47	0.42	0.38	0.55	0.52	0.52	0.56	0.43	0.36	0.42	0.40
	Energy	0.31	0.30	0.33	0.30	0.29	0.30	0.31	0.34	0.37	0.29	0.34	0.38	0.33	0.35	0.40	0.36	0.30	0.31	0.30	0.31

IV. EXPERIMENTS

In this section, we find the effectiveness of our datasets in enhancing three different natural language processing tasks by targeting the following research questions:

- RQ1: Which audio features are most effective for analyzing the emotion and its causes?
- RQ2: How well do multilingual datasets perform in the context of Indian languages and cultural nuances?
- RQ3: How effective are these datasets in improving a task that uses both emotion and cause detection?

A. RQ1: Feature Selection

This Section aims to identify which audio features are most effective in analyzing the underlying causes of emotions for both HinEmo and BenEmo. To explore this, we focused on five key audio features: **pause**, **tone**, **pitch**, and **energy** via Librosa and prosodic **stress** calculated as the product of pitch variation and energy. In this analysis, we adopt the forward selection method on features to iteratively understand what combination of features best derives the causes of an emotion. Our analysis in Table III revealed that among the individual features examined at *Iteration* 1, **Stress** was found to be one of the most effective for classifying emotional causes. Higher stress levels were strongly associated with emotions such as happiness and anger. Tone played a critical role in the *Iteration* 2 that involves combining remaining features with stress. In *Iteration* 3, stress, tone, and pause together were most useful for understanding the causes behind emotions. We intuit that, along with stress and tone, pauses, especially long or frequent ones, were indicative of hesitation or contemplation, often correlating with uncertainty or sadness. The following iterations do not enhance the cause extraction task any further.

B. RQ2: Emotion and Cause Extraction

To evaluate the effectiveness of multilingual multimodal datasets in the context of Indian languages and cultural nuances, we conducted a series of experiments. Each experiment was designed to assess whether the existing datasets could capture the language characteristics in the Indian context by using different input modalities. Results in Table IV after the models were fine-tuned on the respective datasets with a learning rate of 0.005, a batch size of 32, and training for 10 epochs with the Adam optimizer depict that: 1) Existing SOTA transformer models trained on **only textual features** demonstrated reasonable performance across all datasets. 2) **Only Audio-based models** generally outperformed their text-only counterparts, proving better effectiveness in recognizing emotions and causes. 3) The highest accuracies were achieved using the **multimodal approach** (Text + Audio via cross-attention[29]). This setup allowed the model to leverage information from both modalities, leading to more robust classification. In all three modalities, when trained on ECF, the test accuracy on both datasets was notably lower, indicating a domain and language mismatch. Models trained and tested on the respective Indian datasets showed significant

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TABLE IV
COMPARISON RESULTS ON BOTH EMOTION AND CAUSE EXTRACTION TASK USING VARIOUS MODALITIES

		Emotion Detection																Cause Detection															
		Train:**ECF**, Test:**HinEmo**				Train & Test:**HinEmo**				Train:**ECF**, Test:**BenEmo**				Train & Test:**BenEmo**				Train:**ECF**, Test:**HinEmo**				Train & Test:**HinEmo**				Train:**ECF**, Test:**BenEmo**				Train & Test:**BenEmo**			
	Model	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*	*P*	*R*	*F1*	*A*
Only Text	RoBERTa	0.32	0.33	0.32	0.36	0.54	0.55	0.55	**0.60**	0.43	0.41	0.42	0.40	0.65	0.66	0.65	**0.61**	0.35	0.37	0.36	0.45	0.49	0.51	0.50	**0.57**	0.33	0.30	0.31	0.28	0.54	0.48	0.51	**0.45**
	BERT	0.35	0.31	0.33	0.32	0.55	0.50	0.52	**0.47**	0.32	0.34	0.33	0.31	0.53	0.57	0.55	**0.49**	0.38	0.43	0.40	0.32	0.51	0.56	0.53	**0.43**	0.29	0.34	0.31	0.34	0.47	0.54	0.50	**0.51**
	DistilBERT	0.19	0.21	0.20	0.18	0.39	0.36	0.37	**0.33**	0.16	0.29	0.21	0.27	0.34	0.42	0.38	**0.43**	0.30	0.27	0.28	0.18	0.43	0.40	0.41	**0.39**	0.24	0.19	0.21	0.22	0.42	0.36	0.39	**0.37**
	ALBERT	0.36	0.31	0.33	0.28	0.51	0.48	0.49	**0.41**	0.31	0.33	0.32	0.34	0.45	0.46	0.45	**0.48**	0.32	0.38	0.35	0.43	0.48	0.53	0.50	**0.56**	0.31	0.29	0.30	0.32	0.50	0.47	0.48	**0.49**
	XLNet	0.43	0.36	0.39	0.40	0.60	0.58	0.59	**0.59**	0.34	0.32	0.33	0.35	0.56	0.53	0.54	**0.56**	0.40	0.42	0.41	0.49	0.56	0.59	0.57	**0.64**	0.43	0.36	0.39	0.43	0.62	0.58	0.60	**0.57**
	DeBERTa	0.45	0.48	0.46	0.41	0.66	0.70	0.68	**0.72**	0.42	0.38	0.40	0.42	0.68	0.64	0.66	**0.67**	0.52	0.50	0.51	0.48	0.68	0.69	0.68	**0.65**	0.50	0.42	0.46	0.46	0.72	0.65	0.68	**0.68**
Only Audio	Wav2Vec 2.0	0.39	0.37	0.38	0.31	0.61	0.58	0.59	**0.54**	0.33	0.34	0.33	0.36	0.51	0.53	0.52	**0.53**	0.31	0.30	0.30	0.34	0.47	0.45	0.46	**0.50**	0.26	0.30	0.28	0.29	0.43	0.48	0.45	**0.43**
	SpeechT5	0.48	0.41	0.44	0.42	0.69	0.63	0.66	**0.62**	0.37	0.35	0.36	0.43	0.66	0.64	0.65	**0.61**	0.47	0.41	0.44	0.49	0.70	0.66	0.68	**0.65**	0.44	0.44	0.44	0.50	0.62	0.67	0.64	**0.70**
	HuBERT	0.38	0.43	0.40	0.44	0.63	0.66	0.64	**0.65**	0.46	0.42	0.44	0.48	0.73	0.71	0.72	**0.71**	0.46	0.40	0.43	0.52	0.71	0.67	0.69	**0.71**	0.47	0.46	0.46	0.48	0.68	0.70	0.69	**0.69**
	Data2Vec-Audio	0.29	0.30	0.29	0.37	0.54	0.46	0.50	**0.52**	0.29	0.30	0.29	0.33	0.42	0.52	0.46	**0.49**	0.34	0.33	0.33	0.36	0.48	0.47	0.47	**0.54**	0.35	0.31	0.33	0.31	0.56	0.52	0.54	**0.47**
	Whisper	0.31	0.27	0.28	0.34	0.37	0.41	0.39	**0.39**	0.18	0.23	0.20	0.17	0.35	0.41	0.38	**0.38**	0.36	0.31	0.33	0.30	0.49	0.46	0.47	**0.45**	0.25	0.22	0.23	0.28	0.42	0.41	0.41	**0.42**
Multimodal	VATT	0.22	0.22	0.22	0.17	0.36	0.32	0.34	**0.36**	0.22	0.16	0.19	0.25	0.40	0.32	0.36	**0.42**	0.18	0.29	0.22	0.34	0.38	0.44	0.41	**0.51**	0.29	0.24	0.26	0.24	0.46	0.42	0.44	**0.39**
	AudioCLIP	0.26	0.33	0.29	0.28	0.55	048	0.51	**0.47**	0.28	0.37	0.32	0.28	0.49	0.61	0.54	**0.47**	0.16	0.27	0.20	0.21	0.38	0.42	0.40	**0.35**	0.23	0.18	0.20	0.25	0.40	0.37	0.38	**0.41**
	CLAP	0.40	0.41	0.40	0.43	0.69	0.68	0.68	**0.75**	0.40	0.39	0.39	0.41	0.64	0.64	0.64	**0.65**	0.42	0.39	0.40	0.47	0.66	0.62	0.64	**0.62**	0.42	0.43	0.42	0.47	0.63	0.66	0.64	**0.68**
	SpeechBERT	0.49	0.47	0.48	0.47	0.78	0.76	0.77	0.82	0.46	0.48	0.47	0.51	0.75	0.79	0.77	0.83	0.52	0.50	0.51	0.52	0.76	0.75	0.75	0.77	0.52	0.53	0.52	0.53	0.74	0.76	0.75	0.80
	Flamingo	0.33	0.34	0.33	0.35	0.56	0.60	0.58	**0.61**	0.42	0.43	0.42	0.39	0.64	0.66	0.65	**0.60**	0.48	0.46	0.47	0.49	0.72	0.68	0.70	**0.72**	0.46	0.50	0.48	0.48	0.65	0.73	0.69	**0.68**

TABLE V
COMPARISON RESULTS OF RESPONSES GENERATED USING BOTH DATASETS WITH CAUSES OF VARIOUS MODALITIES. *NOTE: FOLLOWING EXISTING WORKS IN EMPATHETIC RESPONSE GENERATION [29], THE METRICS USED ARE* $PPL \leftarrow Perplexity$, $D{-}1, {-}2$AND-3 $\leftarrow Distinct{-}1, {-}2$ AND-3 SCORES

	HINEMO												BENEMO											
	Text Causes + Response				Audio Causes + Response				Multimodal Causes + Response				Text Causes + Response				Audio Causes + Response				Multimodal Causes + Response			
Model	*PPL*	*D-1*	*D-2*	*D-3*	*PPL*	*D-1*	*D-2*	*D-3*	*PPL*	*D-1*	*D-2*	*D-3*	*PPL*	*D-1*	*D-2*	*D-3*	*PPL*	*D-1*	*D-2*	*D-3*	*PPL*	*D-1*	*D-2*	*D-3*
GPT-2 (small)	15.274	0.412	0.574	0.616	15.043	**0.735**	0.527	0.767	**13.598**	0.699	**0.633**	**0.805**	17.609	**0.683**	0.697	0.632	17.288	0.518	**0.796**	0.686	**12.705**	0.468	0.524	**0.747**
(medium)	22.243	0.558	0.533	0.615	22.580	0.473	0.495	0.661	**18.115**	**0.757**	**0.734**	**0.818**	23.853	0.475	0.519	0.597	17.562	**0.639**	0.707	0.672	**11.718**	0.487	**0.780**	**0.736**
(large)	23.585	0.630	0.774	0.635	21.148	0.647	**0.796**	0.717	**18.920**	**0.655**	0.715	**0.763**	20.085	0.515	0.784	0.500	17.984	0.579	**0.824**	0.473	**12.718**	0.686	0.533	**0.618**
DialoGPT (small)	21.034	0.698	0.586	0.607	20.875	**0.784**	**0.714**	0.662	**14.027**	0.520	0.672	**0.837**	23.830	0.633	0.642	0.637	18.671	**0.667**	0.718	0.706	**12.320**	0.462	0.838	**0.714**
(medium)	24.315	0.474	0.595	0.605	21.219	0.557	0.542	0.844	**20.446**	**0.641**	**0.754**	0.892	21.278	**0.632**	**0.675**	0.596	18.823	0.630	0.665	0.615	**16.297**	0.528	0.561	**0.742**
(large)	27.728	**0.619**	0.745	0.695	22.180	0.445	0.550	0.717	**21.711**	0.539	**0.776**	**0.776**	20.919	0.470	0.495	0.665	19.929	**0.613**	**0.811**	0.750	**15.556**	0.452	0.674	**0.763**
GPT-J	31.244	0.697	0.560	0.572	24.989	0.689	**0.615**	0.668	**24.280**	0.880	0.578	**0.732**	20.066	**0.573**	0.683	0.619	16.718	0.556	**0.790**	0.647	**15.513**	0.566	0.670	**0.727**
T5	20.461	0.748	0.770	0.760	14.173	0.566	0.620	0.780	10.420	**0.784**	0.814	**0.845**	23.592	0.627	**0.768**	0.634	12.844	0.513	0.756	0.682	10.286	**0.669**	0.715	0.806

TABLE VI
HUMAN EVALUATION RESULTS USING CRITERIA FROM EXISTING EMPATHETIC RESPONSE GENERATION WORKS [29]

	HINEMO															BENEMO														
Model	*Empathy*			*Relevance*			*Correctness*			*Clarity*			*LLM-as-judge*			*Empathy*			*Relevance*			*Correctness*			*Clarity*			*LLM-as-judge*		
	Text	Audio	Both	Text	Audio	Both	Text	Audio	Both	Text	Audio	Both	T	Au	MM	Text	Audio	Both	Text	Audio	Both	Text	Audio	Both	Text	Audio	Both	Text	Audio	Both
GPT-2 (small)	3.13	3.28	3.83	3.79	3.85	4.00	3.79	3.90	4.26	4.10	4.13	4.23	3.22	3.58	3.49	3.49	3.85	4.01	3.43	3.57	4.26	3.60	3.85	3.95	3.45	3.51	4.50	3.86	3.55	3.21
(medium)	3.22	3.32	3.82	3.55	3.80	4.06	4.06	4.15	4.26	3.84	3.93	4.34	3.34	3.58	4.00	3.23	3.63	3.83	3.47	3.97	4.42	3.54	3.94	4.24	3.31	3.58	4.23	3.61	3.42	3.28
(large)	3.29	3.68	4.00	3.71	3.76	3.87	3.90	4.13	4.11	3.82	3.95	4.36	3.17	3.55	3.98	3.50	3.68	3.77	3.45	3.82	4.23	3.63	3.85	3.97	3.43	3.66	4.29	3.57	3.49	3.39
DialoGPT (small)	3.30	3.79	4.07	3.60	3.73	4.07	4.17	4.22	4.36	3.83	3.84	4.15	3.15	3.43	4.01	3.40	3.49	3.66	3.31	4.09	4.20	3.56	3.93	4.06	3.37	3.67	4.03	3.52	3.49	3.37
(medium)	3.25	3.81	3.97	3.57	3.80	4.19	3.81	4.16	4.16	3.83	4.10	4.30	3.17	3.47	3.63	3.47	3.69	3.79	3.42	3.68	4.43	3.56	3.72	4.18	3.39	3.54	3.82	3.88	3.45	3.18
(large)	3.07	3.89	4.01	3.60	3.99	4.15	3.99	4.16	4.23	3.82	4.09	4.28	3.20	3.51	3.97	3.28	3.45	3.88	3.48	3.81	4.36	3.47	3.94	4.02	3.42	3.63	4.10	4.01	3.60	3.30
GPT-J	3.11	3.60	3.98	3.56	3.72	3.85	3.99	4.03	4.34	3.92	4.06	4.30	3.24	3.47	4.02	3.03	3.21	3.82	3.41	3.83	4.21	3.70	4.20	3.96	3.44	3.79	4.49	4.03	3.56	3.39
T5	3.10	3.51	3.94	3.47	3.55	4.06	3.88	4.11	4.35	3.88	4.29	4.13	3.31	3.43	3.83	3.46	3.69	3.88	3.46	3.87	4.30	3.44	3.94	4.22	3.46	3.65	3.81	3.82	3.45	3.31

improvements, highlighting the importance of culturally and linguistically relevant training data.

C. RQ3: Empathetic Response Generation

In this study, we evaluate the effectiveness of different modalities—text, audio, and their combination—in improving the Empathetic Response Generation (ERG) task. Following[29], we use causes derived from various modalities + conversation history to evaluate generated responses by various decoders using both automatic metrics (perplexity and distinct scores) and human evaluation. For human evaluation, responses are rated based on empathy, relevance, correctness, and clarity by four evaluators excluded from Section III-B to ensure impartiality. As demonstrated in Table V, when we use text-only data to extract emotion causes, the generated responses demonstrate adequate perplexity scores. However, these responses often fail to capture more subtle emotional cues that can be conveyed through prosody or tone. Hence, when audio-only data is used to derive emotion causes, the responses are notably better at capturing the affective state of the speaker. However, sometimes they struggle with generating responses that are contextually relevant. Hence, the most significant improvements in responses are observed when we use a multimodal approach, integrating both text and audio data via a cross-attention mechanism for emotion cause extraction. Additionally, Table VI depicts how human evaluators and LLM-asjudge (ChatGPT) consistently prefer the responses generated by the multimodal cause models, highlighting their ability to generate both emotionally accurate and contextually appropriate responses.

V. CONCLUSION

In this work, we introduce two multimodal datasets tailored to the Indian audience, capturing unique linguistic nuances. On Indian test data, the emotion detection task improves by $5 - 35\%$ in Hindi and $14 - 35\%$ in Bengali. The cause detection is enhanced by $11 - 25\%$ in Hindi and $14 - 27\%$ in Bengali. Lastly, the empathetic response generation task observes significant enhancement through the use of both text and audio. These results highlight the importance of multimodal approaches in addressing the complexities of Indian language and emotion. For future work, expanding the dataset to include more regional languages and exploring real-time, interactive systems could further enhance the models' effectiveness.

ACKNOWLEDGMENT

The authors thank Dr. Dolly Pandya from GMERS Medical College and Hospital, Junagadh, and Mrs. Mausumi Mukherjee from Central Hospital, South Eastern Railway, Kolkata, for their invaluable contribution in this work.

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REFERENCES

[1] S. Sharanyaa, T. J. Mercy, and S. V. G, "Emotion recognition using speech processing," in *Proc. 3rd Int. Conf. Intell. Technol.*, 2023, pp. 1–5.

[2] S. Akinpelu, S. Viriri, and A. Adegun, "An enhanced speech emotion recognition using vision transformer," *Sci. Rep.*, vol. 14, no. 1, 2024, Art. no. 13126.

[3] M. Khan, W. Gueaieb, A. El Saddik, and S. Kwon, "MSER: Multimodal speech emotion recognition using cross-attention with deep fusion," *Expert Syst. Appl.*, vol. 245, 2024, Art. no. 122946.

[4] W. Chen, X. Xing, P. Chen, and X. Xu, "Vesper: A compact and effective pretrained model for speech emotion recognition," *IEEE Trans. Affect. Comput.*, vol. 15, no. 3, pp. 1711–1724, Jul.–Sep. 2024.

[5] F. Wang, Z. Ding, R. Xia, Z. Li, and J. Yu, "Multimodal emotion-cause pair extraction in conversations," *IEEE Trans. Affect. Comput.*, vol. 14, no. 3, pp. 1832–1844, Jul.–Sep. 2023.

[6] H. Ma, J. Yu, F. Wang, H. Cao, and R. Xia, "From extraction to generation: Multimodal emotion-cause pair generation in conversations," *IEEE Trans. Affect. Comput.*, vol. 16, no. 2, pp. 586–597, Apr.–Jun. 2025.

[7] B. Li, H. Fei, F. Li, T.-s. Chua, and D. Ji, "Multimodal emotion-cause pair extraction with holistic interaction and label constraint," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 21, no. 11, Aug. 2024, Art. no. 307.

[8] X. Ju, D. Zhang, J. Li, S. Li, and G. Zhou, "Enhanced generative framework with LLMs for multimodal emotion-cause pair extraction in conversations," *IEEE Trans. Multimedia*, vol. 27, pp. 4924–4935, 2025.

[9] Q. Zhao, L. Zong, B. Xu, X. Zhang, and X. Liu, "Multi-modal deep emotion-cause pair extraction for video corpus," in *Proc. Int. Conf. Pattern Recognit.*, pp. 451–466, 2024.

[10] A. G. A., V. V., and R. S. K., "Multimodal emotion cause pair extraction in conversations using knowledge distillation and large language models," in *Proc. Int. Conf. Comput. Intell. Netw. Syst.*, pp. 1–8, 2024.

[11] C. Busso et al., "Iemocap: Interactive emotional dyadic motion capture database," *Lang. Resour. Eval.*, vol. 42, pp. 335–359, 2008.

[12] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "MELD: A multimodal multi-party dataset for emotion recognition in conversations," in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics*, Jul. 2019, pp. 527–536.

[13] T. Saha, A. Patra, S. Saha, and P. Bhattacharyya, "Towards emotion-aided multi-modal dialogue act classification," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, Jul. 2020, pp. 4361–4372.

[14] J. Zhao et al., "M3ED: Multi-modal multi-scene multi-label emotional dialogue database," in *Proc. 60th Annu. Meeting Assoc. Comput. Linguistics*, May 2022, pp. 5699–5710.

[15] G. Shen, X. Wang, X. Duan, H. Li, and W. Zhu, "Memor: A dataset for multimodal emotion reasoning in videos," in *Proc. 28th ACM Int. Conf. Multimedia,* 2020, pp. 493–502.

[16] H. Jiang, X. Zhang, and J. D. Choi, "Automatic text-based personality recognition on monologues and multiparty dialogues using attentive networks and contextual embeddings," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 10, pp. 13821–13822.

[17] Y. Chen et al., "CPED: A large-scale chinese personalized and emotional dialogue dataset for conversational AI," 2022, *arXiv:2205.14727*.

[18] C. Busso, S. Parthasarathy, A. Burmania, M. AbdelWahab, N. Sadoughi, and E. M. Provost, "MSP-IMPROV: An acted corpus of dyadic interactions to study emotion perception," *IEEE Trans. Affect. Comput.*, vol. 8, no. 1, pp. 67–80, Jan.–Mar. 2017.

[19] E. Parada-Cabaleiro, G. Costantini, A. Batliner, A. Baird, and B. Schuller, "EmoFilm - A multilingual emotional speech corpus," in *Proc. Interspeech*, Hyderabad, India, 2018, pp. 3638–3642.

[20] S. R. Livingstone and F. A. Russo, "The Ryerson audio-visual database of emotional speech and song," 2018.

[21] I. Christop, "nEMO: Dataset of emotional speech in Polish," in *Proc. Int. Conf. Lang. Resour. Eval.*, 2024, pp. 12111–12116 .

[22] JagranJosh, "10 most spoken languages in India," 2024. [Online]. Available: https://www.jagranjosh.com/general-knowledge/most-spoken-languages-in-india-by-number-of-speakers-1541764100-1

[23] M. Junchi et al., "MULTICAUSENET temporal attention for multimodal emotion cause pair extraction," *Sci. Rep.*, vol. 15, 2025, Art. no. 19372.

[24] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 28492–28518.

[25] S. Gupta, P. K. Garg, and S. K. Dandapat, "Tone: A 3-tiered ontology for emotion analysis," *IEEE Access*, vol. 13, pp. 52446–52460, 2025.

[26] Y. Perlitz, A. Gera, M. Shmueli-Scheuer, D. Sheinwald, N. Slonim, and L. Ein-Dor, "Active learning for natural language generation," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, H. Bouamor, J. Pino, and K. Bali, Eds., Singapore, ACL, Dec. 2023, pp. 9862–9877.

[27] B. Settles, "Active learning literature survey," Dept. Comput. Sci., Univ. Wisconsin-Madison, Madison, WI, USA, 2009.

[28] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.

[29] S. Gupta and S. K. Dandapat, "SEEC and CHASE: An emotion-cause pair-oriented approach and conversational dataset with heterogeneous emotions for empathetic response generation," *Know.-Based Syst.*, vol. 280, 2023, Art. no. 111039.

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